



SOC-Task_Week_1

SOC Fundamentals and Security Operations

Organization: CyART

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THEORICAL LEARNING

1. SOC Fundamentals and Operations

A Security Operations Center (SOC) is a centralized team responsible for monitoring, detecting, analyzing, and responding to cybersecurity threats 24/7/365.

1.1. SOC Purpose

Purpose	Description
Proactive Threat Detection	Identifying threats before they cause damage
Incident Response	Containing and eradicating security incidents
Continuous Monitoring	24/7 surveillance of networks and systems
Compliance	Meeting regulatory requirements (GDPR, HIPAA, PCI-DSS)

1.2. SOC Roles

Role	Tier	Key Responsibilities
Tier 1 Analyst	Junior	Alert triage, initial investigation, escalation
Tier 2 Analyst	Intermediate	Deep investigation, incident response
Tier 3 Analyst	Senior	Advanced threat hunting, malware analysis
SOC Manager	Management	Team leadership, reporting, strategy
Threat Hunter	Advanced	Proactive searching for undetected threats

1.3. Key Functions of a SOC

A SOC performs several operational security functions:



➤ **Log Analysis**

Security logs from systems, applications, and network devices are collected and analyzed for suspicious activities.

Examples:

- failed logins
- privilege escalation
- unusual process execution

➤ **Alert Triage**

Security alerts are prioritized based on:

- severity
- impact
- confidence level

This helps analysts focus on critical threats first.

➤ **Incident Investigation**

When suspicious activity is detected, analysts investigate:

- source
- target
- timeline
- attacker behavior

➤ **Threat Intelligence Integration**

Threat intelligence helps SOC teams identify:

- malicious IP addresses
- malicious domains
- malware indicators

This improves detection accuracy.

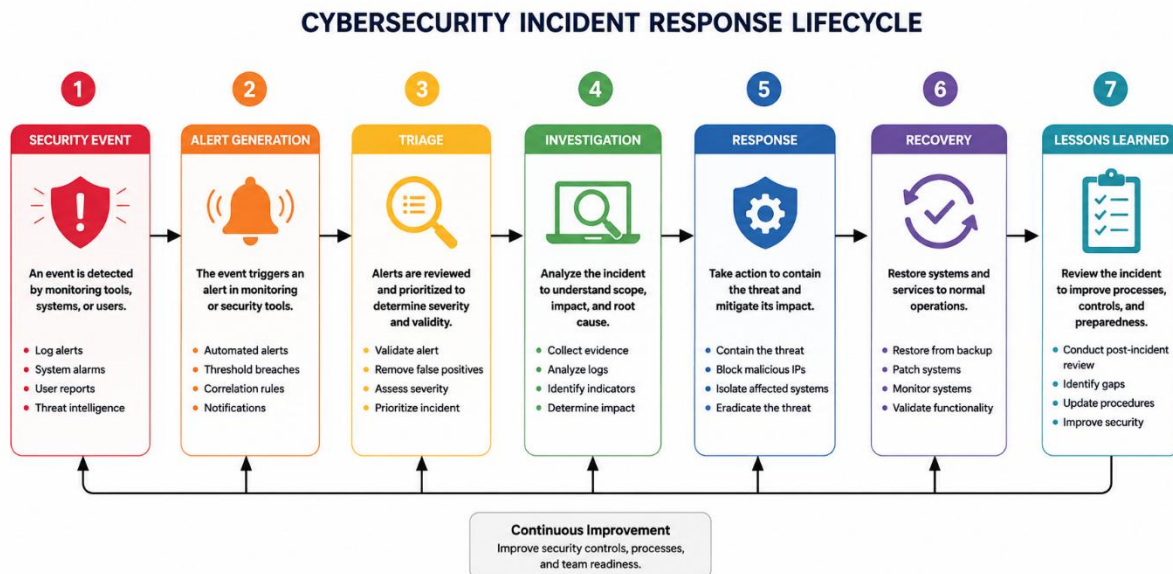
➤ **Threat Hunting**

Threat hunting involves proactively searching for threats that bypass automated detection systems.



This improves security visibility.

1.4. SOC Workflow



1.5. Key Learnings

From this section, the following concepts were learned:

- A SOC is essential for continuous organizational security monitoring.
- SOC operations involve multiple analyst levels with different responsibilities.
- Security monitoring and incident response follow structured workflows.
- Threat hunting helps detect hidden threats before they cause damage.
- Log analysis is a critical component of security operations.

2. Security Monitoring Basics

Security monitoring involves continuously observing network traffic, system logs, and user behavior to detect anomalies, unauthorized access, and policy violations.

2.1. Monitoring Objectives

Objective	Description
Detect Anomalies	Identify deviations from normal behavior
Unauthorized Access	Catch brute force attacks, privilege escalation



Policy Violations	Detect prohibited activities (P2P, data exfiltration)
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2.2. Key Metrics

Metric	Description
False Positives	Benign events incorrectly flagged as threats
False Negatives	Real threats missed by detection systems
MTTD (Mean Time to Detect)	Average time to identify an incident
MTTR (Mean Time to Respond)	Average time to contain/resolve

2.3. Practical Monitoring Activity

A practical traffic analysis exercise was performed using Wireshark to monitor network traffic for approximately 10 minutes.



Wireshark · Protocol Hierarchy Statistics · traffic_10min.pcap

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Pack
▼ Frame	100.0	85488	100.0	135159852	1802 k	0
▼ Ethernet	100.0	85488	0.9	1196832	15 k	0
▼ Internet Protocol Version 6	0.1	54	0.0	2408	32	0
▼ User Datagram Protocol	0.0	23	0.0	184	2	0
Multicast Domain Name System	0.0	23	0.0	1930	25	23
Internet Control Message Protocol v6	0.0	31	0.0	760	10	31
▼ Internet Protocol Version 4	99.9	85370	1.3	1707500	22 k	0
▼ User Datagram Protocol	0.9	746	0.0	5968	79	0
Simple Service Discovery Protocol	0.0	5	0.0	490	6	5
QUIC IETF	0.1	126	0.1	157500	2099	126
NAT Port Mapping Protocol	0.0	3	0.0	26	0	3
Multicast Domain Name System	0.0	23	0.0	1930	25	23
Logical-Link Control	0.0	2	0.0	8	0	0
Local Service Discovery	0.0	4	0.0	476	6	4
Domain Name System	0.4	360	0.0	31801	423	360
Data	0.0	19	0.0	1070	14	19
BitTorrent Tracker	0.0	6	0.0	268	3	6
BitTorrent DHT Protocol	0.2	200	0.0	38391	511	200
▼ Transmission Control Protocol	99.0	84598	1.3	1725180	23 k	67545
Transport Layer Security	19.4	16570	112.9	152546685	2033 k	16570
Data	0.6	483	0.0	19556	260	483
Internet Group Management Protocol	0.0	25	0.0	200	2	25
Internet Control Message Protocol	0.0	1	0.0	36	0	1
Address Resolution Protocol	0.1	64	0.0	1792	23	64
802.1Q Virtual LAN	0.0	5	0.0	20	0	0

No display filter.

Close Copy Protocols Help

3. Log Management Fundamentals

Log management involves collecting, normalizing, storing, and analyzing log data from various sources to detect security incidents.

3.1. Log Lifecycle

Phase	Purpose
Collect	Gather logs from sources (agents, syslog, APIs)
Normalize	Convert to standard format (CEF, JSON)
Store	Index in SIEM for fast searching
Analyze	Query, correlate, alert



Phase	Purpose
Retain	Archive for compliance (30-90 days typical)

3.2. Common Log Types

Different systems generate different types of logs.

➤ Windows Event Logs

Generated by Windows systems.

Examples:

- failed login attempts
- service creation
- account lockouts

Common Event IDs:

Event ID	Description
4625	Failed login attempt
4624	Successful login
7045	New service installed
4688	Process creation

➤ Syslog

Generated by Linux systems and network devices.

Contains:

- authentication logs
- system events
- service logs

Used for centralized monitoring.

➤ Web Server Logs

Generated by web servers.



Examples:

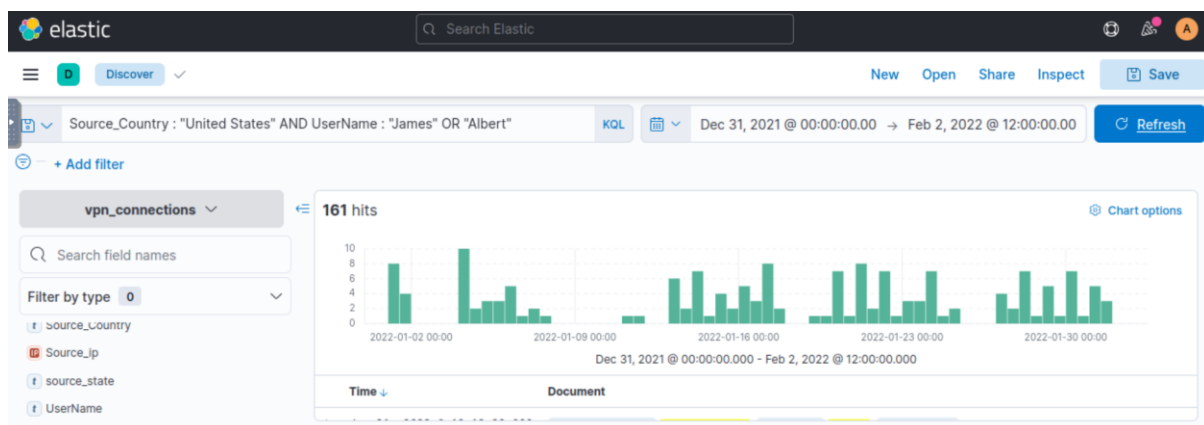
- HTTP requests
- status codes
- client IP addresses

Useful for identifying:

- suspicious requests
- brute-force attempts
- web attacks

3.3. Kubana Query Language (KQL)

In Elastic Stack, KQL was used to filter vpn_connections for Users James or Albert out of the United States



4. Security Tools Overview

Security tools provide specialized capabilities for monitoring, detecting, and responding to threats.

4.1. Security Information and Event Management (SIEM)

Security Information and Event Management (SIEM) systems collect and analyze security logs from multiple sources.

Purpose:

- centralized logging
- alert correlation



- threat detection
- incident investigation

Examples:

- Splunk
- IBM QRadar
- Wazuh

Key learning:

SIEM tools improve visibility by centralizing logs and correlating events.

4.2. Endpoint Detection Response (EDR)

Endpoint Detection and Response (EDR) tools monitor endpoint activities.

Purpose:

- detect malware
- monitor processes
- investigate endpoint behavior

Example:

- CrowdStrike Falcon

Key learning:

EDR improves endpoint visibility and helps detect suspicious activities.

4.3. Intrusion Detection and Prevention Systems (IDS/IPS)

IDS/IPS tools monitor network traffic for malicious activities.

Examples:

- Snort
- Suricata

Purpose:

- detect attacks
- block malicious traffic
- generate alerts

Key learning:

IDS/IPS helps detect attacks at the network level.



4.4. Vulnerability Scanners

Vulnerability scanners identify security weaknesses.

Examples:

- Nessus
- OpenVAS

Purpose:

- identify vulnerabilities
- assess risk
- prioritize remediation

Key learning:

Vulnerability scanning helps proactively identify weaknesses.

4.5. Snort Rule Testing

A custom Snort rule was created to detect HTTP requests to a malicious domain.

```
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on lo, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:08:20.775720 IP 10.255.255.254.37551 > 10.255.255.254.domain: 405+ A? www.msftconnecttest.com. (41)
E..E..@.H.
...
.....5.1.>.....www.msftconnecttest.com.....
16:08:20.843735 IP 10.255.255.254.57161 > 10.255.255.254.domain: 42924+ PTR? 254.255.255.10.in-addr.arpa. (45)
E..I..@.H.
...
.....I.5.5.B.....254.255.255.10.in-addr.arpa.....
16:08:20.983165 IP 10.255.255.254.domain > 10.255.255.254.37551: 405 4/0/0 CNAME www.msftconnecttest.com.edgesuite.net., CNAME a39.d.akamai.net., A 23.52.67.16, A 23.52.67.8 (154)
E.....@.H3
...
.....5.....www.msftconnecttest.com.....'.www.msftconnecttest.com      edgesuite.net..5.....a39.d.akamai.net..h.....4C..h.....4C.
16:08:21.011950 IP 10.255.255.254.domain > 10.255.255.254.57161: 42924 NXDomain 0/0/0 (45)
E..I.. @.H.
...
.....5.I.5.B.....254.255.255.10.in-addr.arpa.....
16:08:21.021326 IP 10.255.255.254.53937 > 10.255.255.254.domain: 40639+ AAAA? www.msftconnecttest.com. (41)
E..E.
@.H.
...
.....5.1.>.....www.msftconnecttest.com.....
16:08:21.195906 IP 10.255.255.254.domain > 10.255.255.254.53937: 40639 2/1/0 CNAME www.msftconnecttest.com.edgesuite.net., CNAME a39.d.akamai.net. (177)
E..../@.G.
...
.....5.....www.msftconnecttest.com.....'.www.msftconnecttest.com      edgesuite.net..5.....a39.d.akamai.W.l.....T...n0d.n
hostmaster.akamai. i.....
^C
6 packets captured
12 packets received by filter
0 packets dropped by kernel
```



```
(dadjonick@Madadjo)~  
$ curl -v http://malicious.com 2>&1 | head -20  
% Total    % Received % Xferd  Average Speed   Time    Time     Time    Current  
         0         0     0         0         0         0         0         0  
* Host malicious.com:80 was resolved.  
* IPv6: (none)  
* IPv4: 64.176.195.8, 45.77.200.164  
* Trying 64.176.195.8:80...  
* Trying 45.77.200.164:80...  
* Established connection to malicious.com (64.176.195.8 port 80) from 172.20.23.5 port 36018  
* using HTTP/1.x  
> GET / HTTP/1.1  
> Host: malicious.com  
> User-Agent: curl/8.18.0  
> Accept: */*  
>  
* Request completely sent off  
< HTTP/1.1 308 Permanent Redirect  
< Connection: close  
< Location: https://malicious.com/  
< Server: Caddy  
< Date: Fri, 08 May 2026 16:55:59 GMT
```

5. Basic Security Concepts

Basic security concepts form the foundation of cybersecurity operations. These principles help security analysts understand how to protect systems, assess risks, and respond to threats effectively.

5.1. The CIA Triad

Principle	Purpose	Example
Confidentiality	Protect data from unauthorized access	Encryption
Integrity	Protect data from unauthorized changes	Hashing
Availability	Ensure systems remain accessible	Backup systems

These three concepts are closely related but different.

5.2. Threat, Vulnerability, and Risk Comparison

Concept	Definition	Example
Threat	Potential danger	Malware attack
Vulnerability	Weakness in a system	Unpatched software



Concept	Definition	Example
Risk	Potential impact of exploitation	Data breach

Understanding this relationship helps prioritize security improvements.

5.3. Defense in Depth

Defense in depth is a security strategy that uses multiple layers of protection. Instead of relying on a single security control, multiple controls work together.

Examples of layers:

- firewalls
- antivirus
- IDS/IPS
- SIEM
- endpoint protection

Security benefit:

If one layer fails, another layer can still provide protection.

Example:

Firewall + antivirus + IDS.

Key learning:

Multiple security layers improve resilience.

5.4. Zero Trust Security Model

Zero Trust is a security model based on the principle:

"Never trust, always verify."

This means no user or device is automatically trusted, even inside the internal network.

Core principles:

- verify identity continuously
- enforce least privilege
- monitor all activities

Security benefit:



Reduces insider threats and unauthorized access.

Example:

Multi-factor authentication for internal systems.

Key learning:

Trust must always be validated.

6. Security Operation Workflow

The SOC workflow standardizes how security alerts are processed from initial detection to final resolution. This ensures consistency, reduces response time, and prevents missed incidents.

6.1. Stages of Security Operations Workflow

The security operations workflow consists of four main stages:

Stage	Description
Detection	Identification of suspicious activity through alerts and monitoring
Triage	Validation and prioritization of alerts
Investigation	Analysis of evidence, logs, and indicators of compromise
Response	Containment, eradication, and recovery actions

7. Incident Response Basics

Incident Response (IR) is the structured approach used to identify, manage, and recover from cybersecurity incidents. The goal of incident response is to minimize damage, reduce recovery time, and prevent future occurrences.

The incident response lifecycle consists of 6 major phases:

Phase	Description
Preparation	Prepare systems, tools, and response plans
Identification	Detect and confirm security incidents



Phase	Description
Containment	Limit the spread of the incident
Eradication	Remove the cause of the incident
Recovery	Restore systems to normal operations
Lessons Learned	Analyze the incident and improve defenses

8. Documentation Standards

Documentation is an essential part of SOC operations and incident response. Proper documentation ensures that incidents are tracked, analyzed, and reviewed accurately.

Security documentation helps teams:

- maintain records
- improve consistency
- support investigations
- improve future response

Documentation is critical for operational efficiency and compliance.

8.1. Types of Security Documentation

SOC teams use several types of documentation.

Document Type	Purpose
Incident Reports	Record security incidents and actions taken
Runbooks	Step-by-step operational guides
SOPs (Standard Operating Procedures)	Standardized response procedures
Post-Mortems	Incident review and lessons learned

PRACTICAL APPLICATION



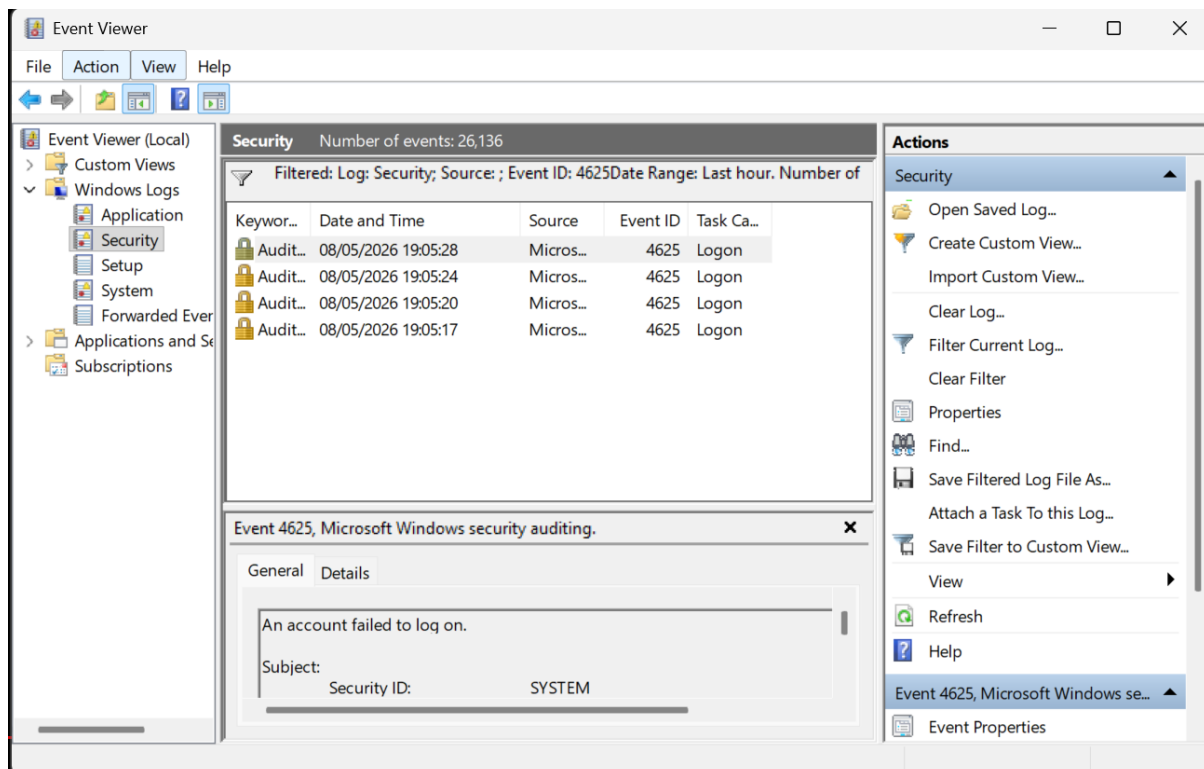
1. Log Analysis Practice

1.1. Windows Event Viewer

Multiple failed login attempts were intentionally generated using incorrect credentials.

Purpose:

Simulate brute-force behavior.





Level	Date and Time	Source	Event ID	Task Category
				Subject: Security ID:SYSTEM Account Name:MADADJO\$ Account Domain:WORKGROUP Logon ID:0x3E7 Logon Type:2 Account For Which Logon Failed: Security ID:NULL SID Account Name:- Account Domain:- Failure Information: Failure Reason:An Error occurred during Logon. Status:0xC000006D Sub Status:0xC0000380 Process Information: Caller Process ID:0xa9c Caller Process Name:C:\Windows\System32\svchost.exe Network Information: Workstation Name:- Source Network Address:127.0.0.1 Source Port:0
Information	08/05/2026 19:05	Microsoft-	4625	Logon
				Subject: Security ID:SYSTEM Account Name:MADADJO\$ Account Domain:WORKGROUP Logon ID:0x3E7 Logon Type:2 Account For Which Logon Failed: Security ID:NULL SID Account Name:-

Event ID 4625 is the primary indicator of failed login attempts. Multiple failures from the same source IP within a short time window (e.g., 5 failures in 60 seconds) strongly indicates a brute-force attack.